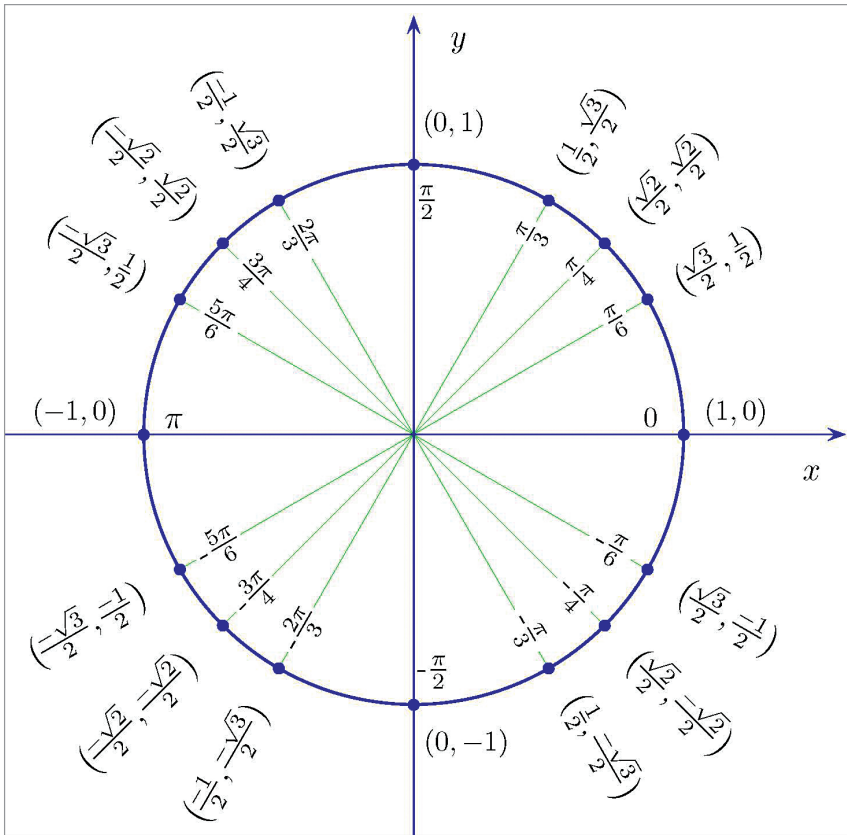




The state of the system where each qubit is represented in the q sphere associated with the computation is represented in the point on the surface of the sphere

Quantum sensors can take advantage of entanglement to enhance their measurement capabilities. For instance, by entangling two particles and measuring one of them, it is possible to gain information about the other particle without directly measuring it. This enables quantum sensors to overcome certain limitations of classical sensors and achieve higher sensitivity.

Quantum sensors employ various techniques to manipulate and measure quantum states. One common technique is using laser light to interact with the quantum system. Laser light interaction with the quantum system allows for precise control and measurement of the sensor's properties. By carefully controlling the laser's properties, such as frequency and intensity, scientists can manipulate the super-

position and entanglement states of the sensor.

Applications

1. Quantum metrology. Revolutionising metrology, quantum sensors enable ultra-precise measurements in atomic clocks, which is essential for global navigation systems like GPS. These clocks exploit the frequency stability of quantum transitions to provide accurate timekeeping.

2. Gravitational wave detection. Quantum sensors play a crucial role in detecting gravitational waves, which are ripples in the fabric of spacetime. Devices like interferometers, which measure minuscule changes in the lengths of two arms, use quantum-enhanced techniques to increase their sensitivity, allowing the detection of incredibly weak gravitational wave signals.

3. Magnetic field sensing. Quantum sensors can measure magnetic fields with exceptional precision. They find applications in diverse areas, such as medical diagnostics, mineral exploration, and environmental monitoring. Magnetic resonance imaging (MRI) machines, for example, utilise quantum sensors to map internal body structures.

4. Quantum communication and cryptography. Quantum sensors are integral to quantum communication systems. They enable secure transmission of information through quantum key distribution (QKD), which uses the principles of quantum mechanics to ensure unbreakable encryption. Quantum sensors can detect eavesdroppers attempting to intercept the transmitted quantum states.

5. Quantum imaging. Quantum sensors enable imaging techniques that surpass classical limitations. Quantum-enhanced imaging can provide higher resolution and sensitivity, making it useful for biomedical imaging, surveillance, and materials characterisation. Quantum ghost imaging, for instance, reconstructs images using correlated photon pairs without directly interacting with the object being imaged.

6. Gravity and inertial sensing. Quantum sensors can measure tiny changes in acceleration and gravitational forces. These sensors have applications in geophysical surveys, inertial navigation systems, and earthquake detection. They can detect slight variations in gravity, enabling the mapping of underground structures or monitoring volcanic activity.

7. Quantum chemical sensing. Quantum sensors enable highly accurate and sensitive detection of chemical compounds. They find applications in environmental monitoring, industrial processes, and medical diagnostics. Quantum